

Anexă la fișa de înregistrare rezultat CDI nr. 4, Dezvoltare tehnologică

Denumire rezultat conform^{1,2,3} cu Fișa de înregistrare rezultat CDI - Dezvoltare tehnologică 4: **Determinarea distribuției ocupării siturilor în nanoparticulele de oxid de fier utilizând spectroscopia Raman**

Se specifică dacă rezultatul este nou sau dacă este un rezultat mai vechi la care s-a introdus o extindere (referire la fișa anterioară). **Some details were published earlier separately, but all the data are reviewed here together for the first time in the form of a comprehensive Method.**

Autori: Fran Nekvapil, Alexander Bunge, Alexandra Falamaș

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Descriere:

1. Introduction

This method file is primarily intended for spectroscopists as a guide in learning to analyse ## Raman spectra of spinel-structured iron oxides with $Fd3m$ symmetry (ferrites). The main intention here is to shorten the learning curve of new generations of spectroscopists by compiling groundwork information on the issues that are fairly clear at the moment of writing.

¹ Rezultatul în sensul considerat aici poate fi atât tangibil (de ex. o clasă nouă de compuși, un material nou, o componentă dintr-un ansamblu care nu exista pe piață etc.), cât și intangibil (de ex. o aplicație / rutină / modul software).

² Mai multe rezultate din aceeași categorie (obținute similar) se raportează împreună ca o clasă de rezultate

³ Dacă rezultatul se bazează pe alte rezultate obținute în institut se face referire la fișa de rezultat înregistrată în baza de date.

This methodology is not complete and all-encompassing, though, and further improvements and modifications can be added once enough experimental data are available to support doing so.

From the point of view of vibrational spectroscopy, two structural units are important: the tetrahedral sites (A-sites) and the octahedral sites (B-sites). Cations at A sites are surrounded by 4 oxygen atoms, hence their Raman signal is expected at higher wavenumbers. On the other hand, metal cations at B sites are surrounded by 6 oxygen atoms, hence their Raman signal is expected at lower wavenumbers. There is twice as much B sites than A sites, and the distribution of metal cations between the two influences the properties of the material, most notably magnetization. Ferrites are a category of metallic nanoparticles with a wide range of applications and are commonly researched in RDI institutions.

In most cases, the task of a Raman spectroscopy expert will be to determine the site occupancy rate in received ferrite nanoparticle (NPs) samples and provide feedback to the chemist performing the synthesis.

The steps to undertake in Raman analysis of spinels:

1. Acquisition of quality Raman spectra
2. Background subtraction
3. Deconvolution and evaluation of fitting correctness
4. Extraction of peaks position, FWHM and area (integrated intensity)
5. Calculation of the site occupancy rate

X-ray diffraction is a commonly used analytical method for determining the lattice properties and strain within NPs, however, Raman spectroscopy is more suitable for determining the distribution of the cation types among the A and B sites. In short, this is because XRD is based on spatial distribution of atoms, and is thus largely insensitive to their identity, while Raman spectroscopy is based on vibrational frequency, which is determined by the size and weight of particular cations.

The current method resulted from in-depth analysis of Raman spectra recorded from magnetite, manganese ferrites, cobalt ferrites, zinc ferrites, nickel ferrites, Mn-Zn ferrites, Co-Zn ferrites and many other supplementary samples. The comprehensiveness of this method stems from the fact that consistent conclusions were drawn from the samples synthesized via different methods, by different chemists, and analysed on more than one spectrometer model. This is a contrast to literature reports which considered limited sets of samples at a time.

The novelty and usefulness of this method lies in providing quantitative information on cations, dopants and vacancies localization, which is difficult to access by other common analytical methods. Furthermore, this method file presents quantitative data on Raman band properties such as FWHM, which is usually not presented in aggregated form in research publications.

Consistent correlation of Raman spectroscopy characterization of ferrite NPs with other common characterisation methods was explored in several of our previous studies (Nekvapil et al., 2020; Nekvapil et al., 2022; Szatmari et al., 2023; Szatmari et al., 2025). The mentioned references and the expanded experimental dataset are sufficient to support the present methodology. The text will also contain helpful notes and observations which could only have been obtained through extensive lab work and spectra processing.

Before passing onto experimental results, some concepts regarding pectroscopy analysis of ferrites have to be clarified.

The atomic interactions within ferrites are predominantly ionic, which are notably weaker and less coherent than covalent bonds. Thus, band positions as well as their widths, will have a large margin of variation, commonly up to 10 cm^{-1} . Thus, the bands

are conventionally identified by their symmetry-derived denomination (A_{1g} , T_{2g} , E_g), rather than their position. This means that, for instance, iron-oxygen interaction will not be referred to as 675 cm^{-1} band, but as tetrahedral Fe-O (within A_{1g}).

Furthermore, the experimental Raman activity (intensity) of certain bands is heavily influenced by the crystalline symmetry of the NPs lattice (White and DeAngelis, 1967). Hence, a common occurrence is that certain bands may be completely inactivated in non-polarized measurements and not recorded at all. This commonly appears with the $T_{2g}(3)$ band (discussed later).

2. Methodology

2.1. Measurement process

2.1.1. Preparation of nanoparticles for analysis

The preparation procedure is the same for all ferrite nanoparticles (NPs) regardless of their origin. The initial sample must be in the form of a dry powder. A small quantity ($< 1\text{ mm}^3$) of dry NPs is placed into a 1 ml capacity glass vial, and covered with 200 μl of distilled water. The vial is subsequently immersed for 30 seconds in an ultrasonic bath to break the NPs aggregates and create a suspension. Using a pipette, 10 μl of the NPs suspension is drop-coated on a microscope slide. A glass slide is acceptable, however, quartz, CaF, or other kind of crystalline or reflective substrate would minimize interference of signal from the substrate with the Raman signal of NPs.

The drop-coating step has to be done fast, as ferrite NPs usually precipitate back to the vial bottom in a matter of minutes. The drop-coated water evaporates in room conditions in 10 to 20 minutes. Using a hot plate for drying should be avoided, as the heat can provoke modifications in the sample structure.

2.1.2. Raman spectroscopy analysis

The NPs are analysed under 632.8 nm excitation. Experiments have shown that shorter wavelength lasers induce local heating and modification of the structure of the nanoparticles, while longer wavelength lasers do not yield enough scattering intensity. Hence, with the 632.8 nm laser, the output power should be set to $< 1\text{ mW}$. If a stronger signal is required, the integration time and number of acquisitions can be increased. If the NPs have higher crystallinity, 60 seconds may be sufficient, while in the case of less crystalline NPs the integration may be as long as 200 seconds.. In this method file, experimental results were acquired using the 100x objective. It is recommended to record at least 3 spectra from each sample, so as to have enough material for spectra interpretation. By chance, deconvolution will work on some spectra better than on others, so it is important to have options.

2.1.3. Interpretation of spectra

Firstly the background has to be subtracted from the acquired spectra, preferably by curved line interpolation between user-defined anchor points. Smoothing is not recommended, nor necessary, however, spectra should be acquired with enough quality so as the shoulders and approximate peak positions can be deduced. Figure 1 shows an example of a background fit to a ferrite spectrum of acceptable quality.

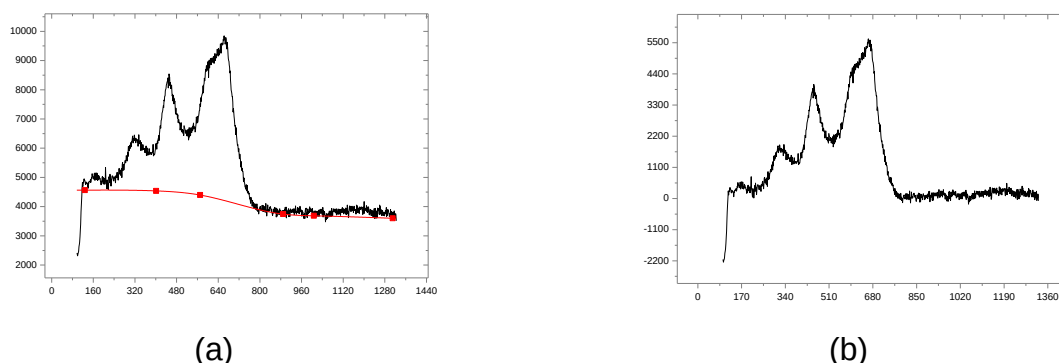


Figure 1. Example of background shape applied to a Raman spectrum of a ferrite: (a): the raw spectrum (black) and the baseline (red); (b) the same spectrum with background subtracted.

Next, a Lorentzian multi-peaks fitting should be applied on the entire spectrum (from ca. 50 to 800 cm^{-1}). This is because the Raman modes of ferrites are broad and significantly overlapping. Exceptional cases will be mentioned later. The number of peaks may range from 5 in magnetite to > 10 in doped ferrites, and is a subject of following an educated guess according to the shape of spectra and expected number of peaks for a particular material.

The sections below will help the analyst determine the number and location of peaks contained within a spectrum. Also, summary tables (Table 2 and Table 3 on page 8) provide the reference values for peaks position and FWHM. Hence, if the peak properties obtained by fitting present big deviations from the reference brackets, the analyst may suspect that the fit is not good. For assessing the correctness of fitting, the R^2 values can additionally be used ($R^2 > 0.95$), and visual inspection of the fitted envelope relative to the experimental spectrum outline must be done.

Important information which the analyst should be communicated include chemical composition of the NPs, synthesis method and upper temperature limit reached during synthesis.

2.2. Baseline case: Raman spectra of ferrites containing only iron

2.2.1. Magnetite (Fe_3O_4)

Raman spectra recorded from magnetite feature 5 Raman-active bands. Figure 2 below shows a typical magnetite spectrum with bands labeled according to their symmetry denominations. These denominations are used in the field literature and are valid for all ferrites. Certain articles argued that all the bands arise from tetrahedral sites (e.g. Thota et al., 2016), however, our and previous experimental results indicate that the

A_{1g} band arises from metal-O vibrations at tetrahedral sites, while the $T_{2g}(2)$ region represents metal-O vibrations at octahedral sites. There is still no consensus on the origin of $T_{2g}(1)$, E_g and $T_{2g}(3)$ regions, so they will not be treated in detail here. In some cases these bands can be so attenuated that deconvoluting them is not feasible.

As a note, a wide dataset of experimental measurements on diverse ferrites show that the intensity and width of the $T_{2g}(2)$ relative to other bands depends strongly on the synthesis method of NPs. For instance, in NPs synthesised by co-precipitation method, the $T_{2g}(2)$ is weak, while in NPs synthesized by sol-gel the band is higher, and can be stronger than A_{1g} .

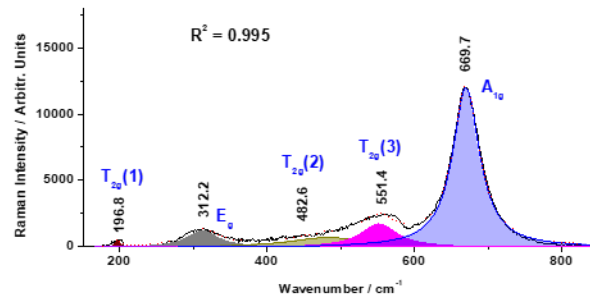


Figure 2. A typical Raman spectrum of magnetite.
Excitation: 632.8 nm (reproduced from Nekvapil et al., 2020).

2.2.2. Maghemite ($\gamma\text{-Fe}_3\text{O}_4$)

Maghemite is a form of spinel-structure iron oxide similar to magnetite, but with reduced magnetization properties arising from lattice defects. The two also exhibit an utterly similar XRD pattern, so it is impossible to differentiate them without high-precision instrumentation and interpretation (Jorgensen et al., 2007), hence Raman spectroscopy is more appropriate for this task. The marker feature of maghemite is a band appearing in 710 – 730 cm^{-1} range (Figure 3). This band is termed here as V_{vacancy} and arises from cation vacancies appearing on partially occupied octahedral sites (Grau-Crespo et al., 2010). Apart from that, the A_{1g} band is present, at 665 cm^{-1} , the $T_{2g}(2)$ is present at 499 cm^{-1} albeit wide, $T_{2g}(3)$ is attenuated, and the E_g is split into two modes. Overall, maghemite spectrum is not difficult to deconvolute as the experimental spectrum outline clearly shows that there should be 6 peaks.

The *Raman defect rate* can be determined semi-quantitatively by calculating the ratio between the areas of V_{vacancy} band around 716 cm^{-1} and Fe-O mode around 665 cm^{-1} :

$$\text{Raman defects rate} = A_{\text{vacancy}} / A_{\text{Fe-O}}$$

Comprehensive discussion of the Raman signal of maghemite is presented by Chourpa et al. (2005). The *Raman defect rate* is a dimensionless unit, and may be either < 1 or > 1 . To date, higher level of defects in ferrites are not desirable, however, their effects are not well explored, except direct linking to saturation magnetization property by Nekvapil et al. (2020).

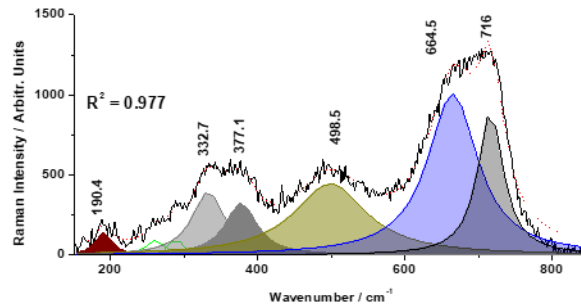


Figure 3. A typical Raman spectrum of maghemite.
Excitation: 632.8 nm (reproduced from Nekvapil et al., 2020).

2.3. Single cation doping of spinel-structure iron oxides

2.3.1. Theoretical localization of cations

When speaking about doping of spinel-structure iron oxides, the new cation may be incorporated in tetrahedral and octahedral sites. A concept of *preference energy for octahedral sites* has been established to determine at which site a particular cation would be in thermodynamic equilibrium (i.e. which site does a particular cation prefer). For the current purpose, Table 1 (from Kovtunenکو, 1997) displays the values for common dopants. In brief, the more negative the value, the higher the probability that the cation will occupy octahedral sites. For instance, Mn^{2+} has no preference, while Mn^{4+} has a strong preference for octahedral sites.

Table 1. Preferred enthalpy (octahedral preference energy) in kJ mole⁻¹ for common cations (reproduced from Kovtunen, 1997).

Cation	Preferred enthalpy, kJ/mole			
	Thermo-dynamic	Quantum-mechanical calculations,* determination after		
		[9]	[10]	[11]
Si ⁴⁺	+ 50.2 ± 4.2	—	—	—
Cu ¹⁺	+ 46	0	0	0
Zn ²⁺	+ 25.1 ± 4	0	0	—
Cr ⁶⁺	+ 5.9	0	0	—
Ga ³⁺	+ 5 ± 0.9	0	0	—
Ag ¹⁺	+ 4.2	0	0	—
Mn ²⁺	- 0 ± 4	0	0	—
Na ¹⁺	- 4	—	—	—
Cd ²⁺	- 20.9	0	0	—
Cr ⁴⁺	- 20.9	—	—	—
Fe ²⁺	- 20.9	- 16.7	- 16.3	- 17
Ge ⁴⁺	- 21.3 ± 2.1	0	0	—
Fe ³⁺	- 22.2 ± 2.9	0	0	0
Mg ²⁺	- 25.1	—	—	—
In ³⁺	- 27.6 ± 6.3	0	0	—
Cr ²⁺	- 29	—	—	—
Co ²⁺	- 29.3 ± 4.2	- 31	- 8.8	- 55
Ti ³⁺	- 37.7	- 28.9	- 32.2	+ 100
V ⁴⁺	- 40.1	—	—	+ 81
Ti ²⁺	- 41.8	—	—	—
Li ¹⁺	- 41.8	—	—	—
Mn ⁴⁺	- 43	—	—	—
V ³⁺	- 43.5	- 53.6	- 50.2	- 266
Al ³⁺	- 44.4 ± 4.2	—	—	—
Ti ⁴⁺	- 46	0	0	0
Ni ²⁺	- 46	- 86.2	- 95.4	- 265
Mn ³⁺	- 50	- 95.4	- 106	—
Sn ⁴⁺	- 59	0	0	—
Cu ²⁺	- 84	- 63.6	- 65.3	—
V ²⁺	- 134	—	- 132	—
Cr ³⁺	- 137 ± 10	- 158	- 195	- 294

As will be seen further, cations such as Zn²⁺, Co²⁺ or Mn²⁺ exhibit their respective metal-O modes at lower wavenumber than iron cations. This red shift arises from a bigger crystalline radius of Zn, Co and Mn relative to Fe. Figure 4 depicts a loose correlation between the crystalline radii of common cations in ferrites with the position where they exhibit their A_{1g} mode. Furthermore, Table 2 and Table 3 present the usual wavenumber positions for Mn-O, Co-O, Zn-O and Fe-O modes at both A and B sites, extracted from our large dataset of experimental spectra on the respective ferrites.

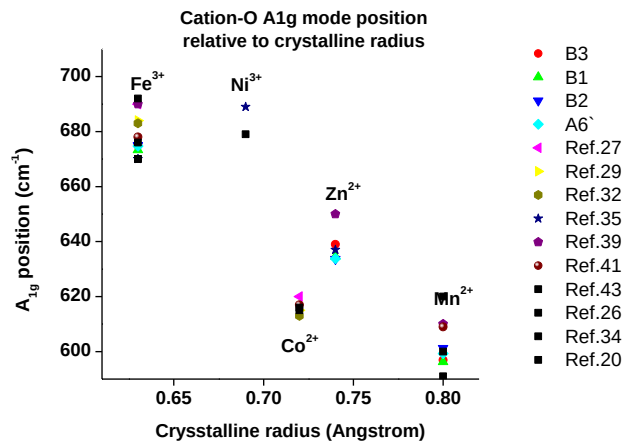


Figure 4. Plot of the metal-O symmetric stretching mode (A_{1g}) relative to the crystalline radius of respective cations. Reproduced from Nekvapil et al. (2020; the references in the legend correspond to the references from the publication).

Table 2. Properties of modes contained in 590 - 730 cm^{-1} region. (all ferrite types).

Band	Mode	Centre	Centre range	FWHM
A_{1g}	Mn-O	599.12 ± 3.48	592 - 609	65.72 ± 13.13
	Co-O	628.50 ± 4.40	616 - 630	92.39 ± 17.87
	Zn-O	640.84 ± 6.93	655 - 631	73.59 ± 18.06
	Fe - O	681.97 ± 7.76	664 - 697	49.23 ± 13.17
	Ni-O	$702 \pm 1,41$	700 - 703	$29,59 \pm 5,33$
V_{vacancy}	defects	$722,54 \pm 4,66$	712 - 730	$34,60 \pm 5,28$

Table 3. Properties of modes contained in 400 - 580 cm^{-1} region (all ferrite types).

Band	Mode	Centre	Centre range	FWHM
$T_{2g}(2)$	Mn-O	441.40	428 - 452	71.06 ± 12.91
	Ni-O	$456,33 \pm 1,15$	455 - 457	$26,96 \pm 2,97$
	Co-O	472.41 ± 2.49	467 - 475	51.94 ± 7.71
	Zn-O	460.73 ± 11.68	454 - 486	66.54 ± 21.27
	Fe-- O	502.58 ± 6.14	487 - 514	75.15 ± 31.05
$T_{2g}(3)$	unassigned	565.14 ± 15.74	544 - 599	74.67 ± 24.34

2.3.2. Manganese ferrites (MnFe_2O_4)

Mn often occurs as a mixture of charged states comprising Mn^{2+} , Mn^{4+} and others. Hence, according to Table 1, Mn atoms may be expected to occupy both A and B sites, depending of their charge. Tetrahedral Mn is revealed by an additional band around 590 – 610 cm^{-1} , next to the Fe-O mode in around 665 – 690 cm^{-1} already discussed in

magnetite. At B sites, the mode around 450 cm^{-1} corresponds to Mn-O, next to a Fe-O mode around $520 - 540\text{ cm}^{-1}$.

Figure 5 below shows the refined results from an experiment involving Mn-ferrite with different levels of Mn doping (10 samples, with Mn doping from 10 to 100 %). Figure 5a shows a Raman spectrum from a 20 % Mn-ferrite, and Figure 5b from a 90% Mn-ferrite. In the case of Mn doping, the divalent manganese cations replace the Fe^{2+} cations, hence the idealized formulae for the two cases would be $\text{Mn}^{2+}_{0.2}\text{Fe}^{2+}_{0.8}\text{Fe}^{3+}_2\text{O}_4$ and $\text{Mn}^{2+}_{0.9}\text{Fe}^{2+}_{0.1}\text{Fe}^{3+}_2\text{O}_4$, respectively.

Since the Fe^{2+} in normal spinels occupy A sites, the doping starts from there, and in 20 % Mn-ferrite the additional Mn-O mode at tetrahedral sites can be seen by the appearance of a band around 606 cm^{-1} . At octahedral sites, the Mn-O and Fe-O modes overlap to a great extent, and only a common $\text{T}_{2g}(2)$ band representing both of them could be fitted, appearing around 466 cm^{-1} . Hence, there is no way to reliably measure Mn occupancy of B sites rate in this particular case.

Mn ferrite should exhibit slightly more complex spectrum than pure iron oxide magnetite, due to additional content of manganese cations in addition to iron and oxygen. The deconvolution procedure will go as follows: for the first run, the maximum possible number of peaks should be attempted, which is 6 (2 components A_{1g} , $\text{T}_{2g}(3)$, $\text{T}_{2g}(2)$ band as mentioned appears to be a single shape, E_g and $\text{T}_{2g}(1)$). If the fit is not correct, gives errors or negative bands, then fitting should be run again with progressively smaller number of peaks until a satisfactory fit is obtained (Table 2 and Table 3). Repeated deconvolution of the same spectrum in an attempt to force peaks is likely to distort fitting results, and a better solution would be deconvoluting another spectrum if available.

Experimentally, we recorded Raman spectra of Mn-ferrite featuring five distinct modes (Figure 5a and b). The important features revealing its structure are the modes of tetrahedral Mn-O vibration around 600 cm^{-1} , tetrahedral Fe-O around 677 cm^{-1} , and a $\text{T}_{2g}(2)$ band representing an overlap of octahedral Mn-O and Fe-O vibrations around 466 cm^{-1} . Weak $\text{T}_{2g}(1)$ and E_g bands around 175 and 320 cm^{-1} , respectively, were also recorded.

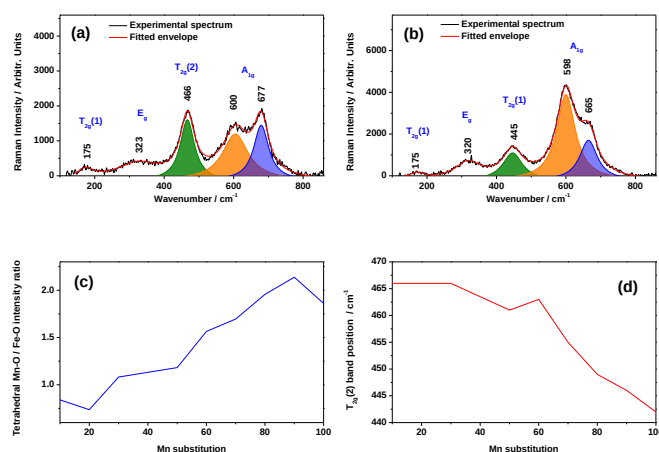


Figure 5. Typical Raman spectra and deconvoluted main bands of Mn substituted ferrites at rates of 0.2 (a) and 0.9 (b), and plots of intensity ratio of 600 and 677 cm^{-1} modes (c) and the wavenumber position of the $\text{T}_{2g}(2)$ band (d) across the samples.

To calculate the *Raman occupancy rate* of Mn at tetrahedral sites, we will consider the intensity ratio of Mn-O band around 500 cm⁻¹ relative to the Fe-O band around 670 cm⁻¹. In this case we use intensity ratio, because the area ratio gives inconsistent results due to slight narrowing of the Mn-O band and slight widening of the Fe-O mode among the samples. Using intensity ratio is also correct, and was used across publications, however it is important to point out that intensity is used rather than area. Hence, the formula says:

$$\text{Mn occupancy rate} = I_{\text{Mn-O}} / I_{\text{Fe-O}};$$

Two trends can be observed as Mn substitution rate increases: (i) the tetrahedral Mn-O band around 600 cm⁻¹ increases in intensity relative to the Fe-O band around 677 cm⁻¹ (Figure 5c), and (ii) the T_{2g}(2) band decreases in intensity and shifts from about 466 cm⁻¹ in low-Mn samples towards 445 cm⁻¹ in highly substituted samples (Figure 5d). In the present case, we may use the intensity ratio of Mn-O and Fe-O modes within the A_{1g} band to quantify the relative localization of Mn to tetrahedral sites. At the lowest Mn substitution rate of 20 %, the intensity ratio was only 0.73, while it climbed to a maximum value of 2.14 at substitution rate of 90 %. The T_{2g}(2) band, representing an overlap of octahedral Mn-O and Fe-O vibrations exhibited a significant shift from 466 cm⁻¹ at Mn substitution rate of 20 % to 445 cm⁻¹ at Mn substitution rate of 90 %. This is indicative of increasing Mn occupancy of both tetrahedral and octahedral sites. The Fe-O band at 677 cm⁻¹ is clearly present in all the samples, meaning that a population of Fe cations always occupies tetrahedral sites.

2.3.3. Zinc (Zn_xFe_{3-x}O₄), cobalt (Co_xFe_{3-x}O₄) and nickel (Ni_xFe_{3-x}O₄) ferrites

Bulk zinc ferrite is almost exclusively a normal spinel, meaning that all the Zn²⁺ cations occupy A sites. However, in nanomaterials and ferrites doped with Zn, this rule may be bent and some zinc can occupy B sites as well. Figure 6 below shows two types of Zn-ferrite. In (a), Zn-ferrite was synthesized by co-precipitation method, while in (b) Zn-ferrite was obtained through complete substitution of Co-ferrite.

Through spectra analysis, it was determined that the best fit for the spectrum in (a) comprises 6 peaks, while the best fit in (b) comprises 5 peaks (because in b the T_{2g}(1) band is flat and was thus exempt from fitting). In both cases we observe contributions of both tetrahedral Zn-O and Fe-O within the A_{1g} band (around 640 and 680 cm⁻¹, respectively), and contributions of both octahedral Zn-O and Fe-O within the T_{2g}(2) band (around 460 and 500 cm⁻¹, respectively).

Raman occupancy rate calculations via the mode area ratios show that in this particular case (Table 5), the sol-gel method yielded Zn-ferrite with higher Zn occupancy rate at A sites and lower at B sites.

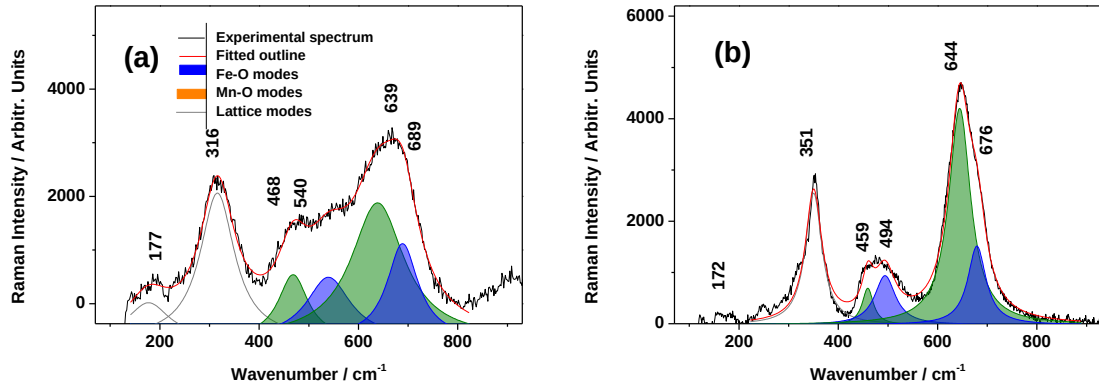


Figure 6. Raman spectra of Zn-ferrite synthesized by two different methods (a) Zn-ferrite synthesized from precursors by co-precipitation method (b) Zn-ferrite obtained by complete substitution of Co-ferrite by sol-gel method.

Table 5. Compact Raman occupancy rate table referring to Figure 5.

Synthesis method	Raman occupancy rate	
	of Zn at B sites	of Zn at A sites
Coprecipitation (a)	0.42	0.71
Sol-gel (b)	0.29	0.77

Cobalt cations exhibit a preference to octahedral sites (Table 1), giving rise to a $T_{2g}(2)$ mode around 475 cm^{-1} , however, at higher substitution levels, it also occupies tetrahedral sites, where it gives rise to an A_{1g} mode around 625 cm^{-1} . The Co-O wavenumbers are close to those of Mn-O, however, it is not likely that these two cations will be present in the same ferrite sample submitted for Raman spectroscopy analysis. In the example below (Figure 7), Co^{2+} seems to have displaced almost all Fe^{3+} from octahedral sites, as the $T_{2g}(2)$ area contains only the Co-O contribution (near perfect Lorentzian fit for a single peak in the $450 - 550 \text{ cm}^{-1}$ region). Hence, here the Raman occupancy rate for Co at B sites would be 1 (which is not so relevant), but it can still be calculated for A sites using the A_{1g} band:

$$\begin{aligned} \text{Raman occupancy rate} &= A_{\text{Co-O}} / (A_{\text{Co-O}} + A_{\text{Fe-O}}) \\ &= 0.71 \end{aligned}$$

Thus, Raman occupancy rate for Co at A sites is 0.71, while the rate for Fe is 0.29. Were the Co doping lower, an additional Fe-O mode would possibly be visible in $490 - 540 \text{ cm}^{-1}$ range and in that case the occupancy rate could have been calculated also for B sites. Apart from these, the $T_{2g}(1)$ band appears at 177 cm^{-1} , the E_g band at 314 cm^{-1} and a weak $T_{2g}(3)$ band is wedged at 554 cm^{-1} .

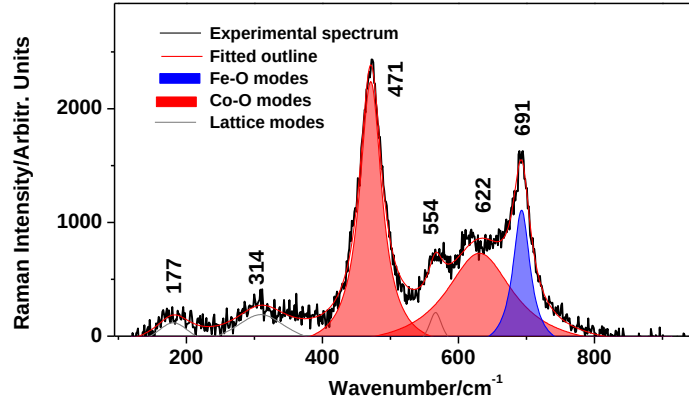


Figure 7. A typical Raman spectrum of Co-ferrite, synthesized by sol-gel method.

Nickel (Ni^{2+}) cations should theoretically abundantly occupy B sites according to their notably negative preference energy (Table 1). However, all of our experimental spectra show a strong A_{1g} component around 703 cm^{-1} and a weak $T_{2g}(2)$ component around 455 cm^{-1} (Figure 8) which would rather indicate localization of Ni^{2+} at A sites. Indeed, the *Raman occupancy rate* of nickel was 0.52 at A sites and 0.18 at B sites for the example in Figure 8, calculated via mode area ratios. Other spectral configuration details include the lack (or weakness) of the $T_{2g}(1)$ band which would appear around $160 - 190 \text{ cm}^{-1}$, the presence of E_g at 333 cm^{-1} and a conspicuous $T_{2g}(3)$ around 577 cm^{-1} . The spectrum is thus deconvoluted with 6 peaks in $300 - 800 \text{ cm}^{-1}$ range. Were the $T_{2g}(1)$ stronger, we may consider 7 peaks in $100 - 800 \text{ cm}^{-1}$ range.

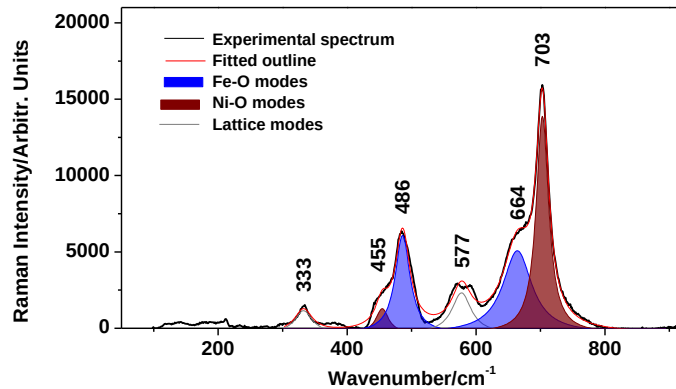


Figure 8. A typical Raman spectrum of Ni-ferrite, synthesized by sol-gel method.

2.4. Three cation ferrite system: MnZnFeO and CoZnFeO

2.4.1. Zinc-doped cobalt ferrite ($\text{Co}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$; $0.1 \leq x \leq 1$)

Combining the knowledge above, we can now tackle the most complicated systems that are likely to be sent to Raman spectroscopy analysis. In this case, we are considering a series of Co-ferrite samples that are progressively doped with Zn, starting

from 10 % until 100 %, and we are going to conduct deconvolution and site occupancy analysis. Firstly, we should enumerate the expected Raman modes:

- Fe-O modes can be expected in all samples, as iron oxide forms the base lattice, hence we will count in one mode within the A_{1g} and one mode within the $T_{2g}(2)$ bands;
- Zn-O is expected only at A sites at lower doping rates, however, at higher doping rates its contribution can be seen also at B sites, hence for Zn we will surely count in one mode within the A_{1g} and possibly another mode within $T_{2g}(2)$ at higher doping rates;
- Co-O is expected to occur at both A and B sites in the lower doping rate samples, however, it may be displaced from A sites by Zn at higher doping rate. Hence, we will definitely consider a Co-O mode within the $T_{2g}(2)$ band, and we will have in mind that there should be a Co-O mode within the A_{1g} band which would progressively decrease in area as the Zn doping increases;
- $T_{2g}(3)$ band may be weak or not visible at all.
- $T_{1g}(1)$ and E_g bands are expected, but may be weak.

Thus, the spectra should be deconvoluted with a minimum of 8 peaks, but as many as 10 are plausible. The exact number will be determined by analysing fitting results (Table 2, Table 3) and the expertise of the analyst.

Figure 9 shows examples of analysis of the respective Raman spectra. Indeed, the octahedral site modes, $T_{2g}(2)$ were deconvoluted into components around 465 – 472 cm^{-1} for Co-O, 495 – 505 cm^{-1} for Fe-O, and 544 – 565 cm^{-1} for Fe-O. The A_{1g} was deconvoluted into components at 613 – 615 cm^{-1} for Co-O, 641 – 655 cm^{-1} for Zn-O, and 680 – 691 cm^{-1} for Fe-O. Note that in Zn-Co ferrite the Fe-O modes were notably red-shifted to higher wavenumbers, and this phenomenon is not explained yet. The $T_{2g}(1)$ band was often very weak and thus could be exempt from the deconvolution range without impact on the rest of the modes. The $T_{2g}(3)$ mode was not revealed in any of the samples, indicating that it was attenuated or symmetry forbidden. Likewise, the E_g band could have been exempt in samples with Zn content < 0.20 (20 % doping).

Note further that the octahedral Zn component was not clearly observed in ferrites with Zn < 0.4 (Figure 10), and that a tetrahedral Co component is not clearly observed for any sample with Zn > 0.4. This can be again explained by the preference energy. Thus, the most valuable information that the Raman spectroscopist can provide to the chemist following this series of samples is that there is an inflection point in site occupancy rate trend at Zn doping between 40 and 50 %.

In the article by Szatmari et al. (2023), the spectra of these samples were fitted under constraints that all Co-O, Zn-O and Fe-O modes have to be present in both the A and the B sites in all samples. This resulted in slightly different quantitative results than presented here. However, as mentioned earlier, there is currently not enough data in the published literature to determine which kind of procedure is more accurate.

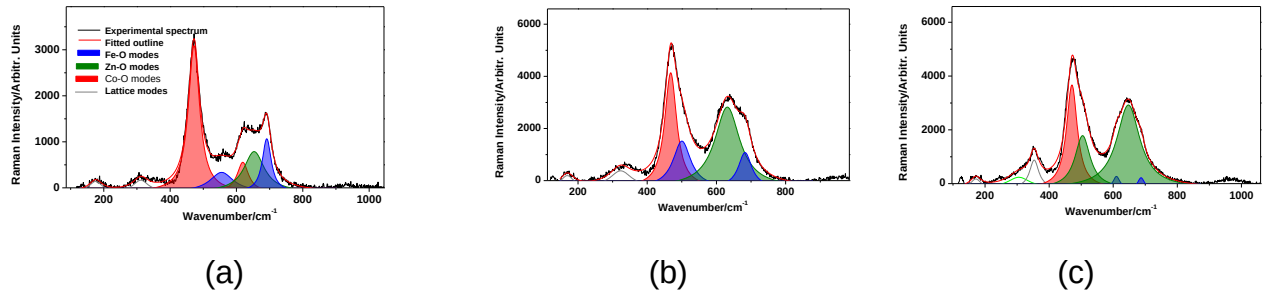


Figure 9. Lorentzian deconvolution of Co-ferrite samples doped with Zn: (a) 20 doping ($\text{Co}_{0.8}\text{Zn}_{0.2}\text{Fe}_3\text{O}_4$); (b) 60 % doping ($\text{Co}_{0.4}\text{Zn}_{0.6}\text{Fe}_3\text{O}_4$); (c) 90 % doping ($\text{Co}_{0.1}\text{Zn}_{0.9}\text{Fe}_3\text{O}_4$).

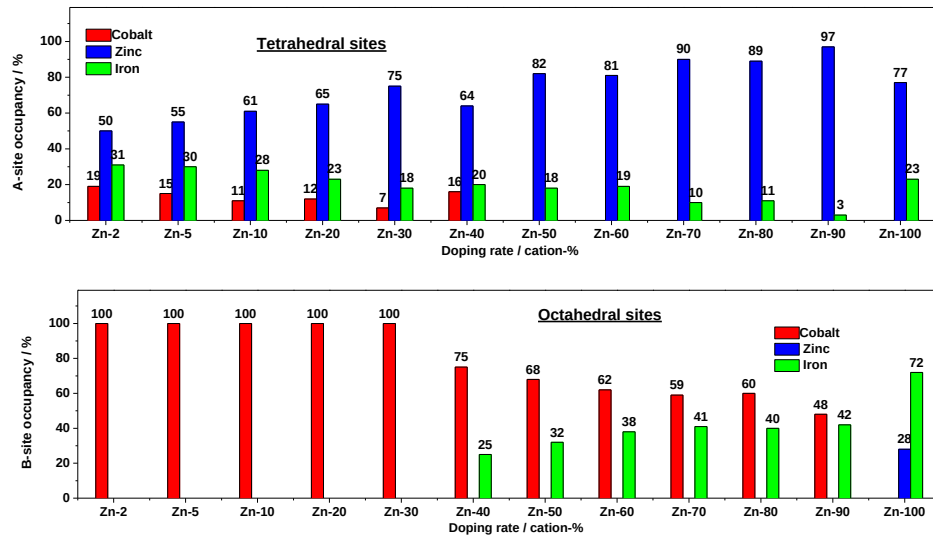


Figure 10. Plot of Raman occupancy rates for tetrahedral sites (modes within the A_{1g} band) and octahedral sites (components of the $T_{2g}(2)$ band) for the Zn-doped Co-ferrite NPs sample.

2.4.2. Manganese zinc ferrite ($\text{Mn}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$: $0.1 \leq x \leq 1$)

When analyzing Raman spectra of manganese zinc ferrite (Mn-Zn ferrite), a similar procedure may be applied as for Zn-doped cobalt ferrite. The sample presented in Figure 11 is a manganese zinc ferrite synthesized by co-precipitation method. Note that previously discussed Co-Zn ferrite was obtained through doping of previously synthesized Co-ferrite with zinc, while this Mn-Zn ferrite was synthesized in a single step. This fact should not change the deconvolution procedure, however, it might influence the relative intensity of the entire $T_{2g}(2)$ envelope, since samples synthesized by sol-gel method usually yield NPs of higher crystallinity than those synthesized by coprecipitation method.

Finally, take into account that NPs synthesized by co-precipitation method may also feature cation vacancies (adding a V_{vacancy} band to the fit), as in Nekvapil et al. (2022). A possible presence of V_{vacancy} band (at $710 - 730 \text{ cm}^{-1}$) can be suspected if the fitted envelope drastically mis-fits the A_{1g} band in its higher wavenumber half (fitting discrepancy at $> 700 \text{ cm}^{-1}$). In this case, one more peak should be added, and its position and FWHM should correspond to the parameters of V_{vacancy} from Table 2.

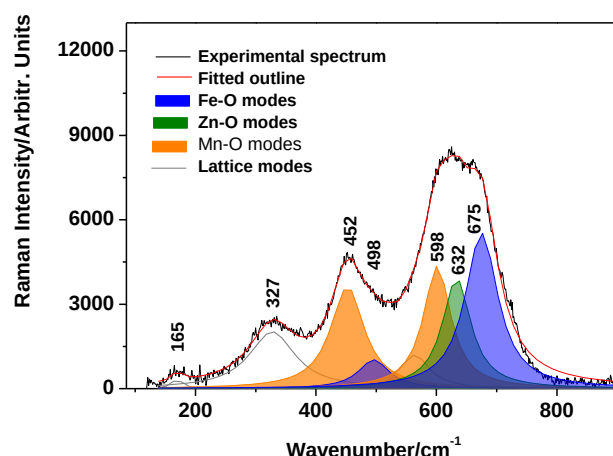


Figure 11. A typical Raman spectrum of Mn-Zn ferrite, synthesized by sol-gel method.

2.5. Validation by Principal Components Analysis

Principal Components Analysis (PCA) was applied on the dataset and the differentiation of the cobalt ferrite NPs doped with various percentages of Zn was obtained. The three-dimensional scatter plot is given in Figure 12. A grouping of the spectra according to the Zn percentage doping can be observed, as a function of the scores of the first three PCs. The total variance of the first three PCs is 83.79% in the dataset. The Zn content increases going from the right side of the image, where the NPs doped with smaller Zn percentages are grouped (20 and 30%), to the left side, where the NPs doped with higher Zn content are clustered. As it can be seen from Fig. 2, the differentiation of the NPs is mainly observed along the first and second PCs.

The first PC presented a variance of 52.4% in the dataset. PC1 presents a positive peak at 471 cm^{-1} and two negative peaks at 353 and 649 cm^{-1} , corresponding in large part to the octahedral Co-O signal. The Raman spectra acquired from the NPs doped with lower Zn percentages, present higher positive values. As the percentage of Zn increases, the associated scores of PC1 decrease reflecting the lower amount of Co remaining in the sample, reaching negative values for the 80-100 % Zn doping.

The second PC, with a variance of 19.3%, shows one main positive peak at 692 cm^{-1} and two negative peaks at 645 and 498 cm^{-1} , referring mostly to tetrahedral Fe-O signal. The scores corresponding to the NPs doped with lower Zn content present higher, positive values. With increasing the Zn content, the associated values of the scores decrease, reaching negative values for the highest Zn content.

The third PC presents a variance of 12% in the dataset and shows negative peaks at 692 and 561 cm^{-1} and positive peaks at 630 and 520 cm^{-1} (representing mostly the tetrahedral Zn-O signal). The scores of PC3 in the spectra increase with increasing the Zn content. Therefore, the Raman spectra acquired from the NPs doped with higher Zn percentage present the positive peaks and the NPs are grouped based on them, while the NPs doped with higher Zn percentage are clustered based on the negative peaks identified in the loading of PC3.

The analysis thus shows that signals of B site Co-O and A site Zn-O are the most relevant elements in analysing the site occupancy rate in Co-Zn ferrites.

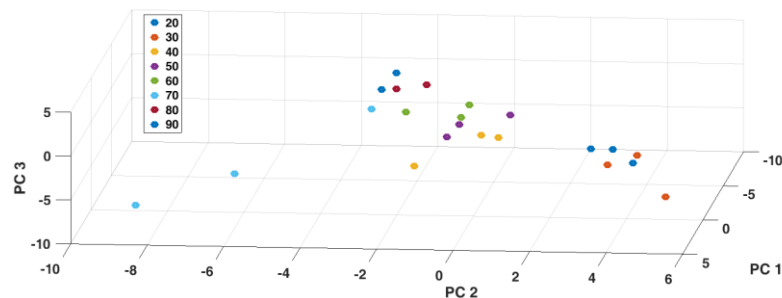


Figure 12. Three-dimensional scatter plot showing the clustering of the cobalt ferrite nanoparticles doped with various Zn content, as a function of the scores of the first three principal components.

As discussed earlier, the presence of cation vacancies is a relevant structural element of Mn-Zn ferrites. However, vacancies are revealed only by a mode V_{vacancy} appearing around 712 to 730 cm^{-1} , which can be quite small compared to the nearby A_{1g} modes and this poses a question about the statistical relevance of their differentiation. PCA was applied on the dataset and the differentiation of the NPs with and without vacancies was obtained, especially, along the scores of PC1. The first PC presented a variance of 95.2% in the dataset. The three-dimensional scatter plot is given in Figure 1.

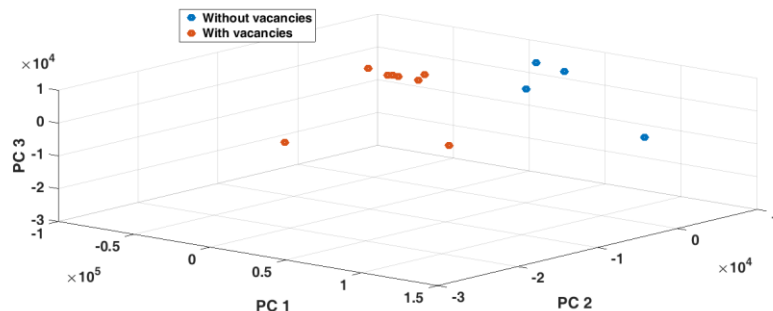


Figure 13. Three-dimensional scatter plot showing the clustering of the Manganese Zinc ferrite nanoparticles with and without vacancies, as a function of the scores of the first three principal components.

The first PC shows peaks located at 662, 618, 454, and 313 cm^{-1} . All of them can be identified in the Raman spectra of the Mn-Zn ferrite nanoparticles without vacancies. Therefore, the associated first principal components' scores of the Raman spectra characteristic to the Mn-Zn ferrite nanoparticles without vacancies are positive, while the scores of the Raman spectra characteristic to the NPs without vacancies are negative. A good separation of the two classes of NPs can be observed along the lines of PC1. Similarly, PC2 also differentiates well between the NPs. Mainly, the NPs without vacancies show larger, positive values, while the NPs with vacancies, present lower, negative values. Although PCA did not show differentiation with respect to the V_{vacancy} , it did show that there are structural differences within the $T_{2g}(2)$ band. These were not explicitly discussed here, but Nekvapil et al. (2020) observed that the presence of vacancies attenuates the Mn-O mode around 428 to 452 cm^{-1} .

3. Valorization prospect

The main application of this method file is to facilitate (routine) quality control of newly synthesized ferrite nanoparticles within INCDTIM, and as a piece of the institute's recorded know-how legacy. This method could be further standardized and formalized in order to create a charged analytical service for economic actors who need qualitative information on ferrites they handle in their operations.

4. References

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